Libertinus and diacritics

Anchor model

Libertinus has an anchor model where the base anchors are generally where the mark would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors.

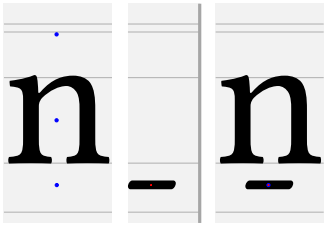

Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is above the mark glyph. In the middle panel of Figure 2, the mark's below-anchor is the red square, above the mark. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors.

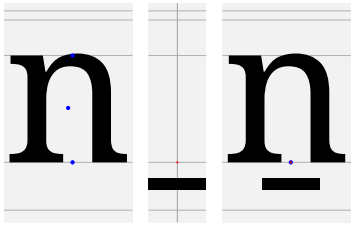

Figure 2: Noto n and macron with anchors.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA vowels for Libertinus serif fonts (regular, italic, semibod, semibold italic) in three tables.

Table 1: Font metrics for A–Z

0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
A ȧ Ạ	B ḃ Ḅ	C ċ Ç	D ḋ Ḍ	E ė Ẹ	F ḟ �f	G ġ Ḡ	H ḣ Ḥ	I İ Ị	J j̇ J	K ḳ Ḳ	L ḷ Ḷ	M ṁ Ṃ	N ṅ Ṇ	O ȯ Ọ	P ṗ Ṗ	Q q̇ Ọ	R ṛ Ṛ	S ṣ Ṣ	T ṭ Ṭ	U u̇ Ụ	V v̇ Ṿ	W ẇ Ẉ	X ẋ Ẋ	Y ẏ Ỵ	Z ż Ẓ
354	267	340	306	281	284	365	361	150	182	336	151	431	346	335	277	348	269	232	307	346	336	450	332	295	320
850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
308	268	363	309	287	209	367	362	147	109	342	262	393	348	350	240		258	234	300	328	318	479	307	295	287
-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
346	280	324	336	271	244	351	362	148	120	323	260	418	353	351	259	359	301	235	297	331	325	474	323	287	309
8	-13.5	16	-30	10	39.5	13.5	-1	2	62	13	-109.5	13	-7	-16.5	17.5	-11	-32	-3.50	10	15	10.5	-24	9	7.50	10.5
0.01	-0.03	0.03	-0.05	0.02	0.09	0.02	-0.00	0.01	0.16	0.02	-0.22	0.02	-0.01	-0.03	0.04	-0.02	-0.06	-0.01	0.02	0.02	0.02	-0.03	0.01	0.01	0.02
-38	-12.5	39	-27	16	-35.5	15.5	0	-1	-11	19	1.50	-25	-5	-1.50	-19.5		-43	-1.50	3	-3	-7.50	5	-16	7.50	-22.5
-0.06	-0.02	0.07	-0.04	0.03	-0.08	0.02	0	-0.00	-0.03	0.03	0.00	-0.03	-0.01	-0.00	-0.04		-0.07	-0.00	0.01	-0.00	-0.01	0.01	-0.02	0.01	-0.04
A ȧ Ạ	*B ḃ Ḅ*	*C ċ Ç*	*D ḋ Ḍ*	*E ė Ẹ*	*F ḟ �f*	*G ġ Ḡ*	*H ḣ Ḥ*	*I İ Ị*	*J j̇ J*	*K ḳ Ḳ*	*L ḷ Ḷ*	*M ṁ Ṃ*	*N ṅ Ṇ*	*O ȯ Ọ*	*P ṗ Ṗ*	*Q q̇ Ọ*	*R ṛ Ṛ*	*S ṣ Ṣ*	*T ṭ Ṭ*	*U u̇ Ụ*	*V v̇ Ṿ*	*W ẇ Ẉ*	*X ẋ Ẋ*	*Y ẏ Ỵ*	*Z ż Ẓ*
505	447	500	396	413	406	513	481	316	392	466	389	557	476	515	426	498	419	402	457	486	481	610	482	435	450
850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
308	238	333	279	269	131	345	320	127	109	292	252	363	328	310	140		248	204	277	308	263	439	287	270	267
-108	-108	-110	-107	-107	-111	-110	-107	-109	-319	-110	-110	-110	-109	-113	-110		-111	-111	-109	-109	-110	-107	-107	-109	-107
334	289	379	355	282	282	395	398	203	263	370	268	445	400	396	295	396	295	272	383	441	418	587	367	393	357
170.5	158	120.5	41	131	123.5	118	83	113	128.5	95.5	120.5	112	76	118.5	131	101.5	124	130	73.5	45	63	23	114.5	42	93
0.26	0.29	0.21	0.06	0.24	0.23	0.19	0.11	0.31	0.28	0.14	0.24	0.13	0.10	0.19	0.23	0.16	0.22	0.29	0.13	0.07	0.10	0.03	0.16	0.08	0.15
-26.5	-51	-46.5	-76	-13	-151.5	-50	-78	-76	-154.5	-78.5	-16.5	-82	-72	-86.5	-155		-47	-68	-106.5	-133	-155	-148	-80.5	-123	-90
-0.04	-0.09	-0.08	-0.11	-0.02	-0.28	-0.08	-0.10	-0.21	-0.33	-0.11	-0.03	-0.10	-0.09	-0.14	-0.28		-0.08	-0.15	-0.20	-0.21	-0.25	-0.16	-0.11	-0.23	-0.14
A ȧ Ạ	**B ḃ Ḅ**	**C ċ Ç**	**D ḋ Ḍ**	**E ė Ẹ**	**F ḟ �f**	**G ġ Ḡ**	**H ḣ Ḥ**	**I İ Ị**	**J j̇ J**	**K ḳ Ḳ**	**L ḷ Ḷ**	**M ṁ Ṃ**	**N ṅ Ṇ**	**O ȯ Ọ**	**P ṗ Ṗ**	**Q q̇ Ọ**	**R ṛ Ṛ**	**S ṣ Ṣ**	**T ṭ Ṭ**	**U u̇ Ụ**	**V v̇ Ṿ**	**W ẇ Ẉ**	**X ẋ Ẋ**	**Y ẏ Ỵ**	**Z ż Ẓ**
									203					363		363	304	252	298		331	493	364	316	321
									806					802		802	808	802	802		805	805	804	805	803
							362		109								292	234	301		303	470	318	316	308
							-110		-319								-111	-111	-107		-108	-107	-107	-109	-107
349	299	332	351	283	268	368	362	160	131	340	267	427	354	365	277	365	320	237	299	339	318	489	354	308	309
									72					-2.50		-2.50	-16	14.5	-1		12.5	4	9.50	7.50	12
									0.17					-0.00		-0.00	-0.03	0.04	-0.00		0.02	0.00	0.01	0.01	0.02
							0		-22								-28	-3.50	2		-15.5	-19	-36.5	7.50	-1
							0		-0.05								-0.05	-0.01	0.00		-0.02	-0.02	-0.05	0.01	-0.00
A ȧ Ạ	***B ḃ Ḅ***	***C ċ Ç***	***D ḋ Ḍ***	***E ė Ẹ***	***F ḟ �f***	***G ġ Ḡ***	***H ḣ Ḥ***	***I İ Ị***	***J j̇ J***	***K ḳ Ḳ***	***L ḷ Ḷ***	***M ṁ Ṃ***	***N ṅ Ṇ***	***O ȯ Ọ***	***P ṗ Ṗ***	***Q q̇ Ọ***	***R ṛ Ṛ***	***S ṣ Ṣ***	***T ṭ Ṭ***	***U u̇ Ụ***	***V v̇ Ṿ***	***W ẇ Ẉ***	***X ẋ Ẋ***	***Y ẏ Ỵ***	***Z ż Ẓ***
354	267	340	306	273	276	365	361	150	432	336	149	431	346	335	316	348	269	232	307	346	336	450	332	295	320
850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
308	268	363	279	279	201	367	365	147	109	342	260	393	348	350	240		258	234	300	328	318	479	307	295	287
-110	-110	-110	-110	-110	-110	-110	-110	-110	-319	-110	-110	-110	-110	-110	-110		-110	-110	-110	-110	-110	-110	-110	-110	-110
364	298	378	367	305	304	405	445	244	288	421	312	449	452	435	307	435	309	263	381	470	417	581	387	407	350
-10.5	-31	-38	-61	-32	-28.5	-40	-84.5	-94.5	144	-85.5	-163	-18.5	-106.5	-100	8.50	-87	-40.5	-31.5	-74.5	-124	-81	-131	-55.5	-112.5	-30.5
-0.01	-0.05	-0.06	-0.09	-0.06	-0.05	-0.06	-0.11	-0.23	0.30	-0.12	-0.31	-0.02	-0.13	-0.15	0.01	-0.13	-0.07	-0.07	-0.13	-0.18	-0.13	-0.14	-0.07	-0.19	-0.05
-56.5	-30	-15	-88	-26	-103.5	-38	-80.5	-97.5	-179	-79.5	-52	-56.5	-104.5	-85	-67.5		-51.5	-29.5	-81.5	-142	-99	-102	-80.5	-112.5	-63.5
-0.08	-0.05	-0.03	-0.13	-0.05	-0.18	-0.06	-0.10	-0.23	-0.37	-0.11	-0.10	-0.07	-0.13	-0.13	-0.12		-0.09	-0.07	-0.14	-0.21	-0.16	-0.11	-0.11	-0.19	-0.10

Table 2: Font metrics for a–z

0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
aȧa̱	bḃḇ	cċc̱	dḋḍ	eėe̱	fḟf̱	gġg	hḣḥ	iiᵢ̱	jjj̱	kk̇ḵ	llḻ	mṁṃ	nṅṇ	oȯo̱	pṗṗ	qq̇q	rṙṛ	sṡṣ	tṫṭ	uu̇u̱	vv̇v̱	wẇw̱	xẋx̱	yẏy̱	zżẕ
209		242	218	243		231	299	137		306	135	392	264	238	271	238	191	206		255	256	345	256	262	220
645		645	645	645		645	645	795		784	885	645	646	645	646	645	646	645		645	646	646	646	645	646
196	234	236	221	234	132	229	272	136	109	290	135	408	265	243	179	348	135	196	163	239	241	352	224	242	207
-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-110	-110	-110	-110	-110	-319	-110
245	231	217	270	222	208	256	270	142	75	261	134	401	275	252	242	268	190	199	163	271	249	373	245	260	220
-36.5		24.5	-52.5	21		-25.5	28.5	-5		45	1	-9.50	-11.5	-14	28.5	-30.5	1	7		-16	7	-28	11	2	-0.50
-0.09		0.07	-0.11	0.06		-0.06	0.06	-0.02		0.09	0.00	-0.01	-0.02	-0.03	0.06	-0.07	0.00	0.02		-0.03	0.01	-0.04	0.02	0.00	-0.00
-49.5	2.50	18.5	-49.5	12	-76.5	-27.5	1.50	-6	33.5	29	1	6.50	-10.5	-9	-63.5	79.5	-55	-3	0	-32	-8	-21	-21	-18	-13.5
-0.12	0.01	0.05	-0.11	0.03	-0.21	-0.06	0.00	-0.03	0.14	0.06	0.00	0.01	-0.02	-0.02	-0.13	0.17	-0.16	-0.01	0	-0.06	-0.02	-0.03	-0.05	-0.04	-0.04
aȧa̱	*bḃḇ*	*cċc̱*	*dḋḍ*	*eėe̱*	*fḟf̱*	*gġg*	*hḣḥ*	*iiᵢ̱*	*jjj̱*	*kk̇ḵ*	*llḻ*	*mṁṃ*	*nṅṇ*	*oȯo̱*	*pṗṗ*	*qq̇q*	*rṙṛ*	*sṡṣ*	*tṫṭ*	*uu̇u̱*	*vv̇v̱*	*wẇw̱*	*xẋx̱*	*yẏy̱*	*zżẕ*
341		343	218	332		360	409			405	292	522	394	359	411	369	312	327		386		348	366	367	360
647		645	645	645		645	645			646	890	645	646	645	646	645	646	645		646		646	646	644	646
174	214	163	221	158	222	197	229	136	0	223	115	382	235	186	248	169	113	169	133	226		355	224	368	187
-108	-108	-113	-107	-108	-111	-319	-111	-107	-319	-107	-107	-105	-111	-107	-110	-110	-110	-111	-107	-111		-110	-110	-107	-111
285	295	253	314	253	222	261	308	202	112	298	202	449	304	271	245	294	259	218	232	311	300	408	285	299	260
56		90	-96	78.5		98.5	100.5			107	90	73	89.5	87.5	166	75	53	109		75		-60.5	80.5	68	99.5
0.14		0.26	-0.21	0.23		0.19	0.23			0.24	0.41	0.11	0.21	0.23	0.31	0.18	0.16	0.34		0.17		-0.09	0.16	0.14	0.24
-111	-81	-90	-93	-95.5	-0.50	-64.5	-79.5	-66	-112	-75	-87	-67	-69.5	-85.5	3	-125	-146	-49	-99.5	-85		-53.5	-61.5	69	-73.5
-0.27	-0.21	-0.26	-0.20	-0.28	-0.00	-0.13	-0.18	-0.35	-0.29	-0.17	-0.40	-0.10	-0.16	-0.22	0.01	-0.29	-0.43	-0.15	-0.38	-0.19		-0.08	-0.12	0.14	-0.18
aȧa̱	**bḃḇ**	**cċc̱**	**dḋḍ**	**eėe̱**	**fḟf̱**	**gġg**	**hḣḥ**	**iiᵢ̱**	**jjj̱**	**kk̇ḵ**	**llḻ**	**mṁṃ**	**nṅṇ**	**oȯo̱**	**pṗṗ**	**qq̇q**	**rṙṛ**	**sṡṣ**	**tṫṭ**	**uu̇u̱**	**vv̇v̱**	**wẇw̱**	**xẋx̱**	**yẏy̱**	**zżẕ**
						231		137			142	396	264							264					
						645		795			885	645	646							646					
						229	277	136	109		135	396	265						163	248					
						-319	-110	-107	-319		-110	-105	-111						-107	-111					
255	235	224	273	232	225	263	275	155	78	261	148	408	289	259	254	273	212	201	168	275	262	384	260	264	221
						-32.5		-18.5			-6.50	-12	-25.5							-11.5					
						-0.07		-0.08			-0.03	-0.02	-0.05							-0.02					
						-34.5	2	-19.5	31		-13.5	-12	-24.5						-5	-27.5					
						-0.08	0.00	-0.08	0.11		-0.05	-0.02	-0.05						-0.02	-0.05					
aȧa̱	***bḃḇ***	***cċc̱***	***dḋḍ***	***eėe̱***	***fḟf̱***	***gġg***	***hḣḥ***	***iiᵢ̱***	***jjj̱***	***kk̇ḵ***	***llḻ***	***mṁṃ***	***nṅṇ***	***oȯo̱***	***pṗṗ***	***qq̇q***	***rṙṛ***	***sṡṣ***	***tṫṭ***	***uu̇u̱***	***vv̇v̱***	***wẇw̱***	***xẋx̱***	***yẏy̱***	***zżẕ***
379		242	218	243		291	469	137		306	174	392	264	238	271	238	191	326		255	256	345		262	220
645		645	645	645		645	645	795		784	890	645	646	645	646	645	646	645		645	646	646		645	646
196	234	236	221	234	132	199	272	136	67	290	135	400	265	243	179	348	135	196	133	239	241	352		242	207
-110	-110	-110	-110	-110	-110	-319	-110	-110	-319	-214	-110	-110	-110	-110	-319	-319	-110	-110	-107	-110	-110	-110		-319	-110
317	303	277	336	277	219	270	338	221	132	339	226	473	344	294	257	313	295	245	259	324	311	418	297	282	267
61.5		-35.5	-118	-34		21	130.5	-84.5		-33.5	-52.5	-81	-80.5	-56.5	13.5	-75	-104.5	80.5		-69.5	-55	-73.5		-20.5	-47.5
0.13		-0.09	-0.22	-0.09		0.04	0.26	-0.38		-0.07	-0.20	-0.11	-0.16	-0.13	0.02	-0.16	-0.25	0.22		-0.14	-0.13	-0.12		-0.04	-0.11
-121.5	-69	-41.5	-115	-43	-87	-71	-66.5	-85.5	-65	-49.5	-91.5	-73	-79.5	-51.5	-78.5	35	-160.5	-49.5	-126.5	-85.5	-70	-66.5		-40.5	-60.5
-0.25	-0.14	-0.10	-0.22	-0.11	-0.14	-0.13	-0.13	-0.38	-0.15	-0.10	-0.35	-0.10	-0.16	-0.12	-0.13	0.07	-0.39	-0.14	-0.40	-0.17	-0.16	-0.11		-0.08	-0.14

Table 3: Font metrics for IPA vowels

0061	0065	0069	006F	0075	0079	00E6	0153	0250	0251	0252	025B	025C	025E	0259	0258	0264	026F	028A	028C	028F	0275	0276
aȧą	eę̇ę	iiḭi̧	oȯǫ	uu̇ṵu̧	yẏy̨y	æǽæ̨	œœ̇œ̨	ɐɐ̇ɐ̨	ɑɑ̇ɑ̨	ɒɒ̇ɒ̨	ɛɛ̇ɛ̨	ɜɜ̇ɜ̨	ɵɵ̇ɵ̨	əə̇ə̨	ɘɘ̇ɘ̨	ɤɤ̇ɤ̨	ɯɯ̇ɯ̨	ʊʊ̇ʊ̨	ʌʌ̇ʌ̨	ʏʏ̇ʏ̨	ɵɵ̇ɵ̨	ɶɶ̇ɶ̨
209	243	137	238	255	262	350	405	228	228	264	209	194	239	195	204	248	407	209	209	209	209	209
645	645	795	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645	645
196	234	136	243	239	242	327		229	212	259	212	184	256	186	212	216	410	258	196	239	196	196
-110	-110	-110	-110	-110	-319	-109		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-110	-107	-110	-110
245	222	142	252	271	260	343	387	216	272	247	201	195	241	232	220	221	390	253	244	236	252	340
-36.5	21	-5	-14	-16	2	6.50	17.5	11.5	-44	17	7.50	-1.50	-2.50	-37	-16.5	27	16.5	-44.5	-35	-27	-43	-131.5
-0.09	0.06	-0.02	-0.03	-0.03	0.00	0.01	0.02	0.03	-0.09	0.04	0.02	-0.00	-0.01	-0.10	-0.04	0.06	0.02	-0.09	-0.07	-0.06	-0.10	-0.21
-49.5	12	-6	-9	-32	-18	-16.5		12.5	-60	12	10.5	-11.5	14.5	-46	-8.50	-5	19.5	4.50	-48	3	-56	-144.5
-0.12	0.03	-0.03	-0.02	-0.06	-0.04	-0.03		0.03	-0.13	0.03	0.03	-0.04	0.03	-0.12	-0.02	-0.01	0.03	0.01	-0.10	0.01	-0.13	-0.23
aȧą	*eę̇ę*	*iiḭi̧*	*oȯǫ*	*uu̇ṵu̧*	*yẏy̨y*	*æǽæ̨*	*œœ̇œ̨*	*ɐɐ̇ɐ̨*	*ɑɑ̇ɑ̨*	*ɒɒ̇ɒ̨*	*ɛɛ̇ɛ̨*	*ɜɜ̇ɜ̨*	*ɵɵ̇ɵ̨*	*əə̇ə̨*	*ɘɘ̇ɘ̨*	*ɤɤ̇ɤ̨*	*ɯɯ̇ɯ̨*	*ʊʊ̇ʊ̨*	*ʌʌ̇ʌ̨*	*ʏʏ̇ʏ̨*	*ɵɵ̇ɵ̨*	*ɶɶ̇ɶ̨*
341	332		359	386	367	480	480				322				204		407					
647	645		645	646	644	645	645				654				645		645					
174	158	136	186	226	368	327		229	212	243	212	184	256	212	212	177	410	258		239		
-108	-108	-107	-107	-111	-107	-109		-114	-114	-114	-114	-114	-114	-114	-114	-114	-114	-114		-107		
285	253	202	271	311	299	394	399	299	297	303	252	228	292	268	260	297	472	312	248	332	289	376
56	78.5		87.5	75	68	85.5	81				69.5				-56		-65.5					
0.14	0.23		0.23	0.17	0.14	0.13	0.13				0.19				-0.16		-0.09					
-111	-95.5	-66	-85.5	-85	69	-67.5		-70.5	-85	-60	-40.5	-44	-36.5	-56	-48	-120	-62.5	-54.5		-93.5		
-0.27	-0.28	-0.35	-0.22	-0.19	0.14	-0.10		-0.17	-0.19	-0.14	-0.11	-0.12	-0.08	-0.16	-0.14	-0.34	-0.09	-0.11		-0.21		
aȧą	**eę̇ę**	**iiḭi̧**	**oȯǫ**	**uu̇ṵu̧**	**yẏy̨y**	**æǽæ̨**	**œœ̇œ̨**	**ɐɐ̇ɐ̨**	**ɑɑ̇ɑ̨**	**ɒɒ̇ɒ̨**	**ɛɛ̇ɛ̨**	**ɜɜ̇ɜ̨**	**ɵɵ̇ɵ̨**	**əə̇ə̨**	**ɘɘ̇ɘ̨**	**ɤɤ̇ɤ̨**	**ɯɯ̇ɯ̨**	**ʊʊ̇ʊ̨**	**ʌʌ̇ʌ̨**	**ʏʏ̇ʏ̨**	**ɵɵ̇ɵ̨**	**ɶɶ̇ɶ̨**
		137		264		355	430				229						377	266				
		795		646		645	645				645						645	645				
		136		248				228						212			380				229	
		-107		-111				-114						-114			-114				-114	
255	232	155	259	275	264	362	401	207	276	251	212	215	249	228	230	202	354	267	231	244	259	334
		-18.5		-11.5		-7.50	28.5				16.5						23	-1				
		-0.08		-0.02		-0.01	0.04				0.05						0.03	-0.00				
		-19.5		-27.5				21						-16.5			26				-30.5	
		-0.08		-0.05				0.05						-0.04			0.03				-0.07	
aȧą	***eę̇ę***	***iiḭi̧***	***oȯǫ***	***uu̇ṵu̧***	***yẏy̨y***	***æǽæ̨***	***œœ̇œ̨***	***ɐɐ̇ɐ̨***	***ɑɑ̇ɑ̨***	***ɒɒ̇ɒ̨***	***ɛɛ̇ɛ̨***	***ɜɜ̇ɜ̨***	***ɵɵ̇ɵ̨***	***əə̇ə̨***	***ɘɘ̇ɘ̨***	***ɤɤ̇ɤ̨***	***ɯɯ̇ɯ̨***	***ʊʊ̇ʊ̨***	***ʌʌ̇ʌ̨***	***ʏʏ̇ʏ̨***	***ɵɵ̇ɵ̨***	***ɶɶ̇ɶ̨***
379	243	137	238	255	262	350	415		228	264	209	194			204		407					
645	645	795	645	645	645	645	645		645	645	645	645			645		645					
196	234	136	243	239	242	327		229	212	259	212	184	256		212	177	410	258		239		
-110	-110	-110	-110	-110	-319	-109		-114	-114	-114	-114	-114	-114		-114	-114	-114	-114		-107		
317	277	221	294	324	282	413	457	290	310	317	265	250	308	278	266	298	467	325	241	341	321	389
61.5	-34	-84.5	-56.5	-69.5	-20.5	-63.5	-42		-82	-53	-56	-56			-62		-60.5					
0.13	-0.09	-0.38	-0.13	-0.14	-0.04	-0.09	-0.06		-0.17	-0.11	-0.14	-0.14			-0.14		-0.08					
-121.5	-43	-85.5	-51.5	-85.5	-40.5	-86.5		-61	-98	-58	-53	-66	-52		-54	-121.5	-57.5	-67		-102.5		
-0.25	-0.11	-0.38	-0.12	-0.17	-0.08	-0.12		-0.14	-0.20	-0.12	-0.13	-0.17	-0.11		-0.12	-0.31	-0.08	-0.12		-0.22		

Regular, above-anchors and below-anchors

See above- and below-anchor coordinates for A–Z and a–z for Libertinus serif regular in Table 4.

Table 4: (X, Y) coordinates of above-anchor (rows 1 and 2) and below-anchor (rows 3 and 4) for A–Z and a–z in Libertinus serif regular.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
354	267	340	306	281	284	365	361	150	182	336	151	431	346	335	277	348	269	232	307	346	336	450	332	295	320
850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850	850
308	268	363	309	287	209	367	362	147	109	342	262	393	348	350	240		258	234	300	328	318	479	307	295	287
−110	−110	−110	−110	−110	−110	−110	−110	−110	−319	−110	−110	−110	−110	−110	−110		−110	−110	−110	−110	−110	−110	−110	−110	−110
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
209		242	218	243		231	299	137		306	135	392	264	238	271	238	191	206		255	256	345	256	262	220
645		645	645	645		645	645	795		784	885	645	646	645	646	645	646	645		646	646	645	646	645	646
196	234	236	221	234	132	229	272	136	109	290	135	408	265	243	179	348	135	196	163	239	241	352	224	242	207
−110	−110	−110	−110	−110	−110	−319	−110	−110	−319	−214	−110	−110	−110	−110	−319	−319	−110	−110	−110	−110	−110	−110	−110	−319	−110

A few bases are missing anchors. The Y coordinates (rows 2 and 4) of base anchors can be set geometrically, relative to a few vertical aspects The X coordinates (rows 1 and 3) of base anchors are better set optically, by judging the optical weight of the top or bottom of the glyph outline, as well as the glyph width.

Vertical spects and clearance

A font has at least five vertical aspects: descender height, baseline, x-height, ascender height, capital height. There are gray horizontal lines of Figure 1, and their Y coordinate is displayed in Table 5. In the anchor model for Libertinus, the Y coordinate of an above-anchor or a below-anchor is at stable value depending on where the glyph outline sits relative to a vertical aspect. For example, all glyphs that sit on the baseline have a below-anchor with $Y = -110$. If a glyph is not a full ascender or full descender, then choose the nearest vertical aspect to the glyph bounding box, and use that vertical aspect's anchor clearance to set the Y coordinate of the anchor. Anchors are often best set optically, not geometrically by this anchor clearance algorithms, and so there may be execptions.

For example, ascender above-anchor clearance is the distance (in absolute value) between the ascender above-anchor Y coordinate and the ascender height, $885 - 698 = 187$.

For a glyph with a descender/ascender beside a counter (bdpqky), the mark attaches sccording to the counter at the x-height or baseline, and for a stem-dominant glyph (fgjlt), the mark attaches according to the major stem at the ascender or descender. Note that an italic f has a descender.

Patch J and Q regular, below-anchor

J's below-anchor is too low. In Table 4, the Y coordinate of the below-anchor of J is the same as that of the lowercase letters with descenders. But the descender of J is not as low as that of j, p, q etc. Look at the J column in the grid labeled regular, Latin, below, which is later in this document. The below-marks are set with too much clearance below J.

Q has no below-anchor (Table 4). By harfbuzz fallback shaping (in the absence of anchors), a below-marks will be place below the baseline, without considering descenders. Look at the J column in the grid labeled regular, Latin, below, which is later in this document. All the below-marks crash into Q's descender.

Let's analyze the below-anchors in order to set the below-anchor for J and Q. The below clearance is clearly 110 units on the Y axis, as can be seen from the below-anchor Y coordinate of the letters that sit on the baseline. What should the below-anchor Y coordinate be for J and Q? For the Y coordinate, the lower bound of the J bounding box is at $Y = -172$. So, the below-anchor Y coordinate of J should arguably be $Y = -172 - 110 = -282$. But the designer set it at $Y = -319$, which is too low. For the Y coordinate, the lower bound of the Q bounding box is at $Y = -209$. So, the below-anchor Y coordinate of Q should arguably be $Y = -209 - 110 = -319$. But the J and Q have curves at the bottoms of their descenders, and to account for overshoot, the anchor should be moved up by a few units. But what is the proper overshoot correction? The O has a baseline overshoot of 10 units. So, the below-anchor Y coordinate of J should really be $Y = -282 + 10 = -272$, and the below-anchor Y coordinate of Q should really be $Y = -319 + 10 = -309$.

Table 5: Vertical aspects and anchor Y coordinates of Libertinus

		Regular	Italic	Semibold	Semibold Italic
height	ascender	698	688–698	690	688–698
	capital	645	645	645	645
	x	429	439, 432–446	433	447, 433–447
	baseline	0	0	0	0
	descender	-232	-238–-229	-238?	-238–-229
above-anchor Y coordinate	ascender	885	885–890	885	885–890
	capital	850	850	850b	850
	x	645	645	645a	645
below-anchor	baseline	-110	-110	-110	-110
	descender	-319	-319	-319	-319
above-anchor clearance	ascender	187	190	195	194
	capital	205	205	205	205
	x-height	216	216	212	198
below-anchor	baseline	110	110	110	110
	descender	87	87	81	100

a top punkt 720
b capitals A–Z 805; other capitals 850

Now, let's decide below-anchor X coordinate of Q. The Q and O are drawn with the same main oval outline coordinates. So, the below-anchor X coordinate of Q should the same as that for O, which is $X = 350$.

That is, the below-anchor of J should be $(109, -272)$, and the below-anchor of Q should be $(350, -309)$. Here is this positioning of J and Q in a patched version of the font (woth O and X for comparison only).

Original: X̯J̯Q̯O̯

Patched: X̯J̯Q̯O̯

But there is no sane reason to put a below-mark on J or Q, so this is not really a problem at all.

Patch k regular, below-anchor, Y coordinate

The below-anchor of k, regular, is at $Y = -214$. But it should be at $Y = -110$, like all other letters that sit on the baseline. That is, the below-anchor of k should be $(109, -110)$.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original: k̯ k̯o ok̯

Patched: k̯ k̯o ok̯

But there is no sane reason to put a below-mark on k, other than perhaps dot (which is a precomposed character), so this is not really a problem at all.

Above anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

Regular, above-anchor, capitals, X coordinate

Figure 3 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 3 for an illustration of increasingly bad (spoofed) X coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

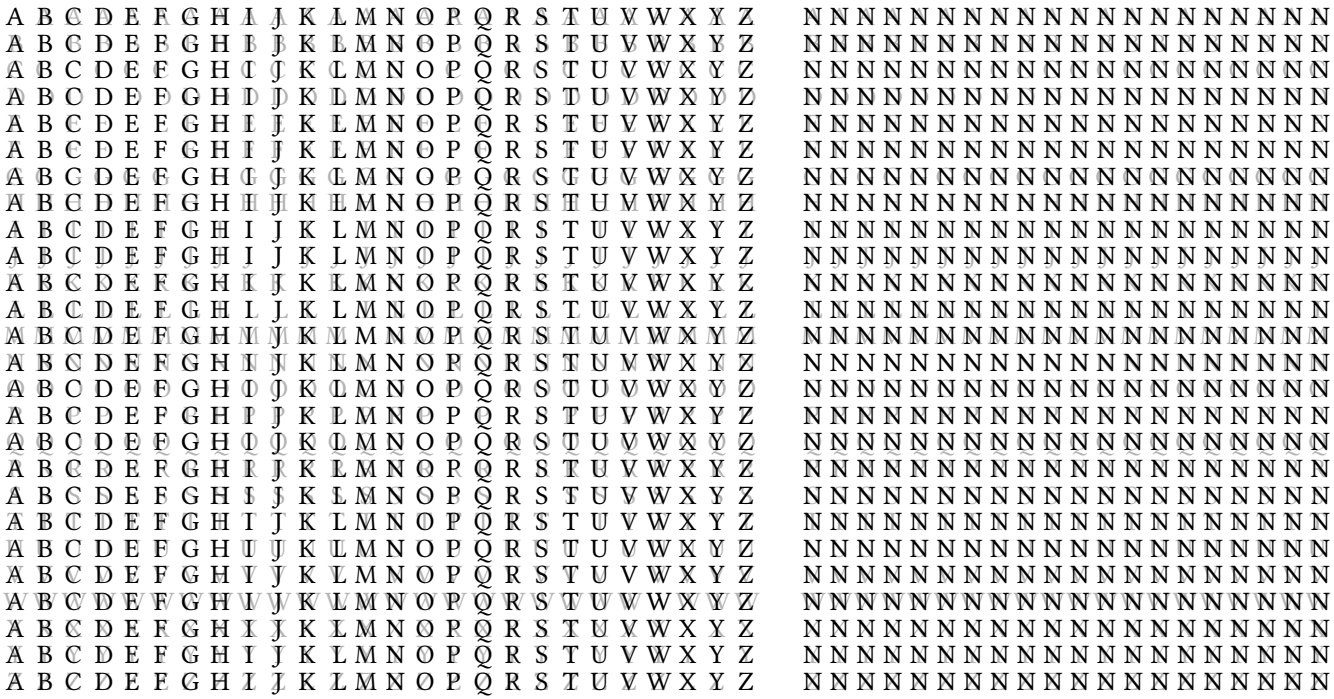

Figure 3: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with N's above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ďďď

`c\char"0327` ç̧ç̧ **ç̧** ***ç̧***
`d\char"0327` ḑḑ **ḑ** ***ḑ***
`e\char"0327` ȩȩ **ȩ** ***ȩ***
`g\char"0327` ġġ **ġ** ***ġ***
`h\char"0327` ḫḫ **ḫ** ***ḫ***
`k\char"0327` ķķ **ķ** ***ķ***
`l\char"0327` ļļ **ļ** ***ļ***
`n\char"0327` ņņ **ņ** ***ņ***
`r\char"0327` ŗŗ **ŗ** ***ŗ***
`s\char"0327` ş̧ş̧ **ş̧** ***ş̧***
`t\char"0327` ţţ **ţ** ***ţ***
This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`.
This behavior is tolerable for the other bases – c, d, e, h, s, t.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.
IJ `\char"0132` U+0132 capital ligature IJ
ij `\char"0133` U+0133 small ligature ij
Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,
`\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}`
ÍJ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute
íj `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute
íj `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute
In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

f with dot below

tiɣə́ ṛf̣a
tiɣə́ ṛf̣a
tiɣə́ ṛf̣a
tiɣə́ ṛf̣a
ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.
f̣ = `f\char"0323` is f with U+0323 combining dot below.
Here are some letters with a dot below.
ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ḟ ġ ṗ ṛ ṭ ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ḡ j ṗ q ẋ
ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ḡ j ṗ q ẋ
ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f ḡ j ṗ q ẋ

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by harfbuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and harfbuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but harfbuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

> For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.
>
> For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə ṛfa	tiɣə ṛfa
tiɣə ṛfa	*tiɣə ṛfa*
tiɣə ṛfa	**tiɣə ṛfa**
tiɣə ṛfa	***tiɣə ṛfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ �w ỵ ẓ ç ḟ ḡ j ṗ q ẋ

Marks

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

Below marks – GPOS anchor 2

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂ x̓ ẍ́ xͅ x͆ x͇ x͈ x͉ x͊ x͋ x͌ x͍ x͎ x͏ x͐ x͑ x͒ x͓ x͔ x͕ x͖ x͗ x͘ x͙ x͚ x͛ x͜ x͝ x͞ x͟ x͠ x͡ x͢

Left angle – GPOS anchor 3

x᜾ x᜾ x᜾ x᜾

Below-right marks – GPOS anchor 4

x̪ x̫ x̬

x̪ x̫ x̬

x̪ x̫ x̬

x̪ x̫ x̬

Cedilla – GPOS anchor 5

x̧ x̧ x̧ x̧

Overlay marks – GPOS anchor 6

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

o͓ o͔ o͕ o͖ o͗

Double marks

$\underbrace{xx}$ $\overbrace{xx}$ $\overline{xx}$ $\underline{xx}$ $\overset{\sim}{xx}$ $\overset{\frown}{xx}$ $\underset{\rightarrow}{xx}$

$\underbrace{xx}$ $\overbrace{xx}$ $\overline{xx}$ $\underline{xx}$ $\overset{\sim}{xx}$ $\overset{\frown}{xx}$ $\underset{\rightarrow}{xx}$

$\check{x}x$ $\acute{x}x$ $\grave{x}x$ $\hat{x}x$ $\breve{x}x$ $\check{x}x$ $\ddot{x}x$

$x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$ $x\boxtimes x$

Overline and low line

O+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines. Custom-width line marks do not have anchors. They are positioned by harfbuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƀƁ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

ƀƆƈƊƋǆƎǝƐƒƔƕƗƘƙƜƝƞƟƢƣƤƥƦƧƨƩƪƫƬƭƮƯưƱƲƳƴƵƶƷƸƹƺƻƼƽƾƿǀǁǂǃǄǅǆǇǈǉǊǋǌǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǝǞǟǠǡǢǣǤǥǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏ

Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

Ớ ơ Ư ư Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̓wkt* [k̓^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̘ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by harfbuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̌t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k̓ʷᵏˣ̌t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k̓ʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ̓ ʱ̓ ʲ̓ ʳ̓ ˢ̓ ᶨ̓ ʷ̓ ʸ̓ ˣ̓ ˡ̓ ˢ̓ ˣ̓ ʿ̓ ᵇ̓ ᵏ̓ ᵐ̓ ᵖ̓ ᵗ̓ ᶬ̓ ᵛ̓ ⱱ̓ ᵝ̓ ᵞ̓ ᵟ̓

ʰ̒ ʱ̒ ʲ̒ ʳ̒ ˢ̒ ᶨ̒ ʷ̒ ʸ̒ ˣ̒ ˡ̒ ˢ̒ ˣ̒ ʿ̒ ᵇ̒ ᵏ̒ ᵐ̒ ᵖ̒ ᵗ̒ ᶬ̒ ᵛ̒ ⱱ̒ ᵝ̒ ᵞ̒ ᵟ̒

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ̓ ʱ̓ ʲ̓ ʳ̓ ˢ̓ ᶨ̓ ʷ̓ ʸ̓ ˣ̓ ˡ̓ ˢ̓ ˣ̓ ʿ̓ ᵇ̓ ᵏ̓ ᵐ̓ ᵖ̓ ᵗ̓ ᶬ̓ ᵛ̓ ⱱ̓ ᵝ̓ ᵞ̓ ᵟ̓

ʰ̒ ʱ̒ ʲ̒ ʳ̒ ˢ̒ ᶨ̒ ʷ̒ ʸ̒ ˣ̒ ˡ̒ ˢ̒ ˣ̒ ʿ̒ ᵇ̒ ᵏ̒ ᵐ̒ ᵖ̒ ᵗ̒ ᶬ̒ ᵛ̒ ⱱ̒ ᵝ̒ ᵞ̒ ᵟ̒

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k̓ʷᵏˣ̒t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſÐđĦħıȷŁłƀƁ

IPA

ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟꞵ
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□
ɒɔɕɖɗɑɤɜɘɛɜɞɟʃɡɠɣɤɥɦɧɨɪɭɬɮɭɫɱɲɳɴɵœɶɸɹɻʀʁʂʃʈʧʊʋʌɯʏɤʐʑʒʔʕʃʗʘʙɕʜʝʞʟ□

Superscript consonants

ʰʱʲʳʴʵʶʷʸˡˢˣˤˢˤˢˤˢ

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẦẤẪẨẰẮẴẲẠẬẶẢẨẲẸỆỂỄỀẾỂỂỄỆỊỈỈĨỊỈĨỈỌỘỖỔỒỐỖỔỒỐỘỎỔỔỖỔỜỚỠỞỢỎỜỚỠỞỢỦỤỤŨ
ŨŪŬŮŰŨŮŰŮŮŮŮŮŮŮŲŲŲŲŪŪŲŮŮŮŮŮŲ

*ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẦẤẪẨẰẮẴẲẠẬẶẢẨẲẸỆỂỄỀẾỂỂỄỆỊỈỈĨỊỈĨỈỌỘỖỔỒỐỖỔỒỐỘỎỔỔỖỔỜỚỠỞỢỎỜỚỠỞỢỦỤỤŨ
ŨŪŬŮŰŨŮŰŮŮŮŮŮŮŮŲŲŲŲŪŪŲŮŮŮŮŮŲ*

**ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẦẤẪẨẰẮẴẲẠẬẶẢẨẲẸỆỂỄỀẾỂỂỄỆỊỈỈĨỊỈĨỈỌỘỖỔỒỐỖỔỒỐỘỎỔỔỖỔỜỚỠỞỢỎỜỚỠỞỢỦỤ
ŨŪŬŮŰŨŮŰŮŮŮŮŮŮŮŲŲŲŲŪŪŲŮŮŮŮŮŲ**

***ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẦẤẪẨẰẮẴẲẠẬẶẢẨẲẸỆỂỄỀẾỂỂỄỆỊỈỈĨỊỈĨỈỌỘỖỔỒỐỖỔỒỐỘỎỔỔỖỔỜỚỠỞỢỎỜỚỠỞỢỦỤ
ŨŪŬŮŰŨŮŰŮŮŮŮŮŮŮŲŲŲŲŪŪŲŮŮŮŮŮŲ***

Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃầấẫẩằắẵẳạậặảẩẳẹệểễềếểểễệịỉỉĩịỉĩỉọộỗổồốỗổồốộỏổổỗổờớỡởợỏờớỡởợủụụũūŭůűũůűũůùùûũūŭůűũůűũůùùûũūŭůűũůűũůùùû

No marks: rare, digraph etc.

Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
Ƹ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ

Precomposed characters of anchor-relevant bases and marks

à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ
à á â ã ā å ą ặ ằ ắ ẳ ấ ầ ẫ è é ê ẽ ē ę ệ ế ề ễ ì í î ĩ ī ị ò ó ô õ ō ọ ộ ố ồ ỗ ù ú û ũ ū ụ ỳ ý ỷ ỹ ȳ ỵ ḍ ḷ ṇ ṛ ṣ ṭ ẓ

Base and mark shaping

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. Note that LSP uses the Libertinus fonts named “Semibold“, e.g. `LibertinusSerif-Semibold.otf`, for all boldface style.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means harfbuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A green color means that a GSUB substitution (to dotless i or j) was expected but did not occur.

A faded color – gray, light blue, pink, or light green – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This on happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o̽ (U+0333 double macron below), o̪ (U+033A inverted bridge below), and o̺ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

There are separate grids for common marks (for IPA etc.) and rare marks (for UPA etc.). Each grid considers all possible bases for marks in that grid, and so the grid is actually a complete matrix or grid or table, but with faded colors indicating combinations that can be ignored because they are not expected in notation in linguistics.

Regular, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɡ ì j ƙ l̀ m̀ ǹ ò ƥ q̀ ɾ̀ s̀ t̀ ù ʋ̀ ẁ x̀ ỳ ƶ æ̀ œ̀ ŋ̀ ɛ̀ ɑ̀ ɔ̀ ɓ̀ ɗ̀ ɗ̀ ə̀ ə̀ ɛ̀ ɜ̀ ɛ̀ ɜ̀ ɡ̀ ɡ̀ ɡ̀ ɣ̀ ɣ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɭ̀ ɮ̀ ʉ̀ ɯ̀ ɲ̀ ɳ̀ ɴ̀ ɵ̀ œ̀ ɪ̀ ɪ̀ ɻ̀ ʀ̀ ʁ̀ ʂ̀ ʃ̀ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ʝ̀ ɻ̀

á ƀ ć đ é f ɡ́ í j ƙ ĺ ḿ ń ó ƥ q́ ŕ ś ƭ ú ʋ́ ẃ x́ ý ź ǽ œ́ ŋ́ ɛ́ ɑ́ ɔ́ ɓ́ ɗ́ ɗ́ ə́ ə́ ɛ́ ɜ́ ɛ́ ɜ́ ɡ́ ɡ́ ɡ́ ɣ́ ɣ́ ɥ́ ɦ́ ɨ́ ɨ́ ɭ́ ɮ́ ʉ́ ɯ́ ɲ́ ɳ́ ɴ́ ɵ́ œ́ ɪ́ ɪ́ ɻ́ ʀ́ ʁ́ ʂ́ ʃ́ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ʝ́ ɻ́

â ƀ ĉ đ ê f ɡ̂ î ĵ ƙ l̂ m̂ n̂ ô ƥ q̂ r̂ ŝ ƭ û ʋ̂ ŵ x̂ ŷ ẑ æ̂ œ̂ ŋ̂ ɛ̂ ɑ̂ ɔ̂ ɓ̂ ɗ̂ ɗ̂ ə̂ ə̂ ɛ̂ ɜ̂ ɛ̂ ɜ̂ ɡ̂ ɡ̂ ɡ̂ ɣ̂ ɣ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɭ̂ ɮ̂ ʉ̂ ɯ̂ ɲ̂ ɳ̂ ɴ̂ ɵ̂ œ̂ ɪ̂ ɪ̂ ɻ̂ ʀ̂ ʁ̂ ʂ̂ ʃ̂ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ʝ̂ ɻ̂

ã ƀ c̃ đ ẽ f ɡ̃ ĩ j ƙ l̃ m̃ ñ õ ƥ q̃ r̃ s̃ ƭ ũ ʋ̃ w̃ x̃ ỹ z̃ æ̃ œ̃ ŋ̃ ɛ̃ ɑ̃ ɔ̃ ɓ̃ ɗ̃ ɗ̃ ə̃ ə̃ ɛ̃ ɜ̃ ɛ̃ ɜ̃ ɡ̃ ɡ̃ ɡ̃ ɣ̃ ɣ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɭ̃ ɮ̃ ʉ̃ ɯ̃ ɲ̃ ɳ̃ ɴ̃ ɵ̃ œ̃ ɪ̃ ɪ̃ ɻ̃ ʀ̃ ʁ̃ ʂ̃ ʃ̃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ʝ̃ ɻ̃

ā ƀ c̄ đ ē f ɡ̄ ī j ƙ l̄ m̄ n̄ ō ƥ q̄ r̄ s̄ ƭ ū ʋ̄ w̄ x̄ ȳ z̄ ǣ œ̄ ŋ̄ ɛ̄ ɑ̄ ɔ̄ ɓ̄ ɗ̄ ɗ̄ ə̄ ə̄ ɛ̄ ɜ̄ ɛ̄ ɜ̄ ɡ̄ ɡ̄ ɡ̄ ɣ̄ ɣ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɭ̄ ɮ̄ ʉ̄ ɯ̄ ɲ̄ ɳ̄ ɴ̄ ɵ̄ œ̄ ɪ̄ ɪ̄ ɻ̄ ʀ̄ ʁ̄ ʂ̄ ʃ̄ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ʝ̄ ɻ̄

å ƀ c̊ đ ɛ̊ f ɡ̊ ɨ̊ j ƙ l̊ m̊ n̊ o̊ ƥ q̊ r̊ s̊ ƭ ů ʋ̊ ẘ x̊ ẙ z̊ æ̊ œ̊ ŋ̊ ɛ̊ ɑ̊ ɔ̊ ɓ̊ ɗ̊ ɗ̊ ə̊ ə̊ ɛ̊ ɜ̊ ɛ̊ ɜ̊ ɡ̊ ɡ̊ ɡ̊ ɣ̊ ɣ̊ ɥ̊ ɦ̊ ɨ̊ ɨ̊ ɭ̊ ɮ̊ ʉ̊ ɯ̊ ɲ̊ ɳ̊ ɴ̊ ɵ̊ œ̊ ɪ̊ ɪ̊ ɻ̊ ʀ̊ ʁ̊ ʂ̊ ʃ̊ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ʝ̊ ɻ̊

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

ą ƀ ç ɖ ę f ɡ ɨ j ƙ l ɱ n o ƥ q ɾ s ƭ u ʋ w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɗ ɗ ə ə ɛ ɜ ɛ ɜ ɡ ɡ ɡ ɣ ɣ ɥ ɦ ɨ ɨ ɭ ɮ ʉ ɯ ɲ ɳ ɴ ɵ œ ɪ ɪ ɻ ʀ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

Regular patched, Bases requiring anchors for common marks, Common combining marks

à ȁ ̀ ̀ ̀

Italic, Bases requiring anchors for common marks, Common combining marks

à ɓ ɕ ɖ è f ɡ ì ɟ ɦ ɨ̀ ɨ̀ ɲ ɔ̀ ɸ ɋ̀ ɾ̀ ʃ̀ ʈ ù ʋ̀ ʍ̀ ɤ̀ ʏ̀ ʒ̀ æ̀ œ̀ ŋ̀ ɐ̀ ɑ̀ ɒ̀ ɓ̀ ɖ̀ ɗ̀ ɘ̀ ə̀ ɛ̀ ɜ̀ ɞ̀ ɟ̀ ɠ̀ ɡ̀ ɢ̀ ɣ̀ ɤ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɭ̀ ɮ̀ ɯ̀ ɱ̀ ɲ̀ ŋ̀ ɴ̀ ɵ̀ ɶ̀ ɹ̀ ɻ̀ ɽ̀ ʁ̀ ʃ̀ ʂ̀ ʊ̀ ʌ̀ ʎ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ɟ̀ ɭ̀

á ɓ ɕ ɗ é f ɡ í ɟ ɦ ɨ́ ɨ́ ɲ ɔ́ ɸ ɋ́ ɾ́ ʃ́ ʈ ú ʋ́ ʍ́ ɤ́ ʏ́ ʒ́ ǽ œ́ ŋ́ ɐ́ ɑ́ ɒ́ ɓ́ ɖ́ ɗ́ ɘ́ ə́ ɛ́ ɜ́ ɞ́ ɟ́ ɠ́ ɡ́ ɢ́ ɣ́ ɤ́ ɥ́ ɦ́ ɨ́ ɨ́ ɭ́ ɮ́ ɯ́ ɱ́ ɲ́ ŋ́ ɴ́ ɵ́ ɶ́ ɹ́ ɻ́ ɽ́ ʁ́ ʃ́ ʂ́ ʊ́ ʌ́ ʎ́ ʏ́ ʐ́ ʑ́ ʒ́ ɟ́ ɭ́

â ɓ ɕ ɗ ê f ɡ î ɟ ɦ ɨ̂ ɨ̂ ɲ ɔ̂ ɸ ɋ̂ ɾ̂ ʃ̂ ʈ û ʋ̂ ʍ̂ ɤ̂ ʏ̂ ʒ̂ æ̂ œ̂ ŋ̂ ɐ̂ ɑ̂ ɒ̂ ɓ̂ ɖ̂ ɗ̂ ɘ̂ ə̂ ɛ̂ ɜ̂ ɞ̂ ɟ̂ ɠ̂ ɡ̂ ɢ̂ ɣ̂ ɤ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɭ̂ ɮ̂ ɯ̂ ɱ̂ ɲ̂ ŋ̂ ɴ̂ ɵ̂ ɶ̂ ɹ̂ ɻ̂ ɽ̂ ʁ̂ ʃ̂ ʂ̂ ʊ̂ ʌ̂ ʎ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ɟ̂ ɭ̂

ã ɓ ɕ ɗ ẽ f ɡ ĩ ɟ ɦ ɨ̃ ɨ̃ ɲ ɔ̃ ɸ ɋ̃ ɾ̃ ʃ̃ ʈ ũ ʋ̃ ʍ̃ ɤ̃ ʏ̃ ʒ̃ æ̃ œ̃ ŋ̃ ɐ̃ ɑ̃ ɒ̃ ɓ̃ ɖ̃ ɗ̃ ɘ̃ ə̃ ɛ̃ ɜ̃ ɞ̃ ɟ̃ ɠ̃ ɡ̃ ɢ̃ ɣ̃ ɤ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɭ̃ ɮ̃ ɯ̃ ɱ̃ ɲ̃ ŋ̃ ɴ̃ ɵ̃ ɶ̃ ɹ̃ ɻ̃ ɽ̃ ʁ̃ ʃ̃ ʂ̃ ʊ̃ ʌ̃ ʎ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ɟ̃ ɭ̃

ā ɓ ɕ ɗ ē f ɡ ī ɟ ɦ ɨ̄ ɨ̄ ɲ ɔ̄ ɸ ɋ̄ ɾ̄ ʃ̄ ʈ ū ʋ̄ ʍ̄ ɤ̄ ʏ̄ ʒ̄ ǣ œ̄ ŋ̄ ɐ̄ ɑ̄ ɒ̄ ɓ̄ ɖ̄ ɗ̄ ɘ̄ ə̄ ɛ̄ ɜ̄ ɞ̄ ɟ̄ ɠ̄ ɡ̄ ɢ̄ ɣ̄ ɤ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɭ̄ ɮ̄ ɯ̄ ɱ̄ ɲ̄ ŋ̄ ɴ̄ ɵ̄ ɶ̄ ɹ̄ ɻ̄ ɽ̄ ʁ̄ ʃ̄ ʂ̄ ʊ̄ ʌ̄ ʎ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ɟ̄ ɭ̄

å ɓ ɕ ɗ ɛ̊ f ɡ̊ ɪ̊ ɟ̊ ɦ̊ ɨ̊ ɨ̊ ɲ̊ ɔ̊ ɸ̊ ɋ̊ ɾ̊ ʃ̊ ʈ̊ ů ʋ̊ ʍ̊ ɤ̊ ʏ̊ ʒ̊ æ̊ œ̊ ŋ̊ ɐ̊ ɑ̊ ɒ̊ ɓ̊ ɖ̊ ɗ̊ ɘ̊ ə̊ ɛ̊ ɜ̊ ɞ̊ ɟ̊ ɠ̊ ɡ̊ ɢ̊ ɣ̊ ɤ̊ ɥ̊ ɦ̊ ɨ̊ ɨ̊ ɭ̊ ɮ̊ ɯ̊ ɱ̊ ɲ̊ ŋ̊ ɴ̊ ɵ̊ ɶ̊ ɹ̊ ɻ̊ ɽ̊ ʁ̊ ʃ̊ ʂ̊ ʊ̊ ʌ̊ ʎ̊ ʏ̊ ʐ̊ ʑ̊ ʒ̊ ɟ̊ ɭ̊

ą ɓ ç ɖ ę f ɡ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ ɓ ç ɖ ę f ɡ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

ɑ̨ ɓ̨ ç ɖ̨ ę f ɡ̨ ɨ̨ ɟ̨ ɦ̨ ɨ̨ ɲ̨ ɔ̨ ɸ̨ ɋ̨ ɾ̨ ʃ̨ ʈ̨ ů ʋ̨ ʍ̨ ɤ̨ ʏ̨ ʒ̨ æ̨ œ̨ ŋ̨ ɐ̨ ɑ̨ ɒ̨ ɓ̨ ɖ̨ ɗ̨ ɘ̨ ə̨ ɛ̨ ɜ̨ ɞ̨ ɟ̨ ɠ̨ ɡ̨ ɢ̨ ɣ̨ ɤ̨ ɥ̨ ɦ̨ ɨ̨ ɨ̨ ɭ̨ ɮ̨ ɯ̨ ɱ̨ ɲ̨ ŋ̨ ɴ̨ ɵ̨ ɶ̨ ɹ̨ ɻ̨ ɽ̨ ʁ̨ ʃ̨ ʂ̨ ʊ̨ ʌ̨ ʎ̨ ʏ̨ ʐ̨ ʑ̨ ʒ̨ ɟ̨ ɭ̨

Italic patched, Bases requiring anchors for common marks, Common combining marks

Semibold, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɠ ì ɟ ƙ ɭ ɱ ɲ ò ƥ ɋ ɾ ʃ ƭ ù ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɻ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɻ

á ƀ ć d é f ɡ í ɟ ƙ ĺ ḿ ń ó ƥ ɋ ŕ ś ƭ ú ʋ ɯ ɤ ý ź ǽ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɽ

â ƀ ĉ d ê f ɡ î ɟ ƙ ĺ ḿ ń ô ƥ ɋ ŕ ś ƭ û ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ã ƀ ɕ d ẽ f ɡ ĩ ɟ ƙ ĺ ḿ ń õ ƥ ɋ ŕ ś ƭ ũ ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɭ

ā ƀ ɕ d ē f ɡ ī ɟ ƙ ĺ ḿ ń ō ƥ ɋ ŕ ś ƭ ū ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɽ

å ƀ ɕ d ɛ f ɡ ɨ ɟ ƙ l ɱ ɲ ɔ ƥ ɋ ŕ ś ƭ ů ʋ ɯ ɤ ý ź æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɭ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

ą ƀ ç d ę f ɡ ɨ ɟ ƙ ɫ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ ɤ ɣ ƶ æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɔ ɓ ɖ ɗ ɵ ə ɛ ɜ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɬ ɮ ɺ ɽ ʉ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɫ ɭ ɾ ʁ ʂ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ɫ

Semibold patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ƈ ɖ è f ɠ ì ɟ ƙ ǀ ɱ ɲ ò ƥ ɋ ɾ ʃ ʈ ù ʋ ɯ x ʏ ʒ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɠ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɭ ɮ ɯ ɱ ɲ ɳ ɴ ɵ œ ɪ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʐ ʑ ʒ ʝ ʟ

á ƃ ć d é f ɡ í ɟ ƙ ǁ ɱ ń ó ƥ ɋ ɾ ś ʈ ú ʋ ɯ x ʏ ź æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʑ ʒ ʝ ɮ

â ƃ ĉ d ê f ɡ î ɟ ƙ ǁ ɱ ɲ ô ƥ ɋ ɾ ŝ ʈ û ʋ ɯ x ʏ ẑ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʑ ʒ ʝ ɮ

ã ƃ c̃ d ẽ f ɡ ĩ ɟ ƙ ǁ ɱ ñ õ ƥ ɋ ɾ s̃ ʈ ũ ʋ ɯ x ʏ z̃ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ā ƃ c̄ d ē f ɡ ī ɟ ƙ ǁ ɱ n̄ ō ƥ ɋ ɾ s̄ ʈ ū ʋ ɯ x ʏ z̄ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

å ƃ c̊ d ɛ̊ f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ɔ̊ ƥ ɋ ɾ ʃ ʈ ů ʋ ɯ x ʏ ʒ æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

ą ƃ ç d ę f ɡ ɨ ɟ ƙ ǀ ɱ ɲ ǫ ƥ ɋ ɾ ʃ ʈ ʉ ʋ ɯ x ʏ z ɐ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɛ ɑ ɒ ɓ ɖ ɗ ɘ ə ɛ ɜ ɵ ɟ ɡ ɡ ɣ ɤ ɥ ɦ ɬ ɬ ɮ ɯ ɱ ɲ ɴ ɵ œ ɪ ɪ ɾ ɹ ʃ ʊ ʌ ʎ ʏ ʒ ʝ ɮ

Semibold italic, Bases requiring anchors for common marks, Common combining marks

Semibold italic patched, Bases requiring anchors for common marks, Common combining marks

à ƀ ɔ̀ đ̀ è f̀ ǵ ì j̀ k̀ l̀ m̀ ǹ ò p̀ q̀ ŕ s̀ t̀ ù v̀ ẁ x̀ ỳ ź æ̀ œ̀ ŋ̀ ɐ̀ à ɒ̀ ɓ̀ ɖ̀ ɗ̀ ə̀ ə̀ ɛ̀ ɜ̀ ɘ̀ ɟ̀ ɡ̀ ɡ̀ ɢ̀ ɣ̀ ɤ̀ ɥ̀ ɦ̀ ɨ̀ ɨ̀ ɩ̀ ɩ̀ ɪ̀ ɪ̀ ɭ̀ ɲ̀ ɴ̀ ɵ̀ œ̀ ɾ̀ ɾ̀ ʂ̀ ʃ̀ ʊ̀ ʌ̀ ʏ̀ ʏ̀ ʐ̀ ʑ̀ ʒ̀ ʝ̀ ɮ̀
á ƀ ɔ́ đ́ é f́ ǵ í j́ ḱ ĺ ḿ ń ó ṕ q́ ŕ ś ť ú v́ ẃ x́ ý ź ǽ œ́ ŋ́ ɐ́ á ɒ́ ɓ́ ɖ́ ɗ́ ə́ ə́ ɛ́ ɜ́ ɘ́ ɟ́ ɡ́ ɡ́ ɢ́ ɣ́ ɤ́ ɥ́ ɦ́ ɨ́ ɨ́ ɩ́ ɩ́ ɪ́ ɪ́ ɭ́ ɲ́ ɴ́ ɵ́ œ́ ɾ́ ɾ́ ʂ́ ʃ́ ʊ́ ʌ́ ʏ́ ʏ́ ʐ́ ʑ́ ʒ́ ʝ́ ɮ́
â ƀ ɔ̂ đ̂ ê f̂ ǵ î ĵ k̂ l̂ m̂ n̂ ô p̂ q̂ r̂ ŝ t̂ û v̂ ŵ x̂ ŷ ẑ æ̂ œ̂ ŋ̂ ɐ̂ â ɒ̂ ɓ̂ ɖ̂ ɗ̂ ə̂ ə̂ ɛ̂ ɜ̂ ɘ̂ ɟ̂ ɡ̂ ɡ̂ ɢ̂ ɣ̂ ɤ̂ ɥ̂ ɦ̂ ɨ̂ ɨ̂ ɩ̂ ɩ̂ ɪ̂ ɪ̂ ɭ̂ ɲ̂ ɴ̂ ɵ̂ œ̂ ɾ̂ ɾ̂ ʂ̂ ʃ̂ ʊ̂ ʌ̂ ʏ̂ ʏ̂ ʐ̂ ʑ̂ ʒ̂ ʝ̂ ɮ̂
ã ƀ ɔ̃ đ̃ ẽ f̃ g̃ ĩ j̃ k̃ l̃ m̃ ñ õ p̃ q̃ r̃ s̃ t̃ ũ ṽ w̃ x̃ ỹ z̃ æ̃ œ̃ ŋ̃ ɐ̃ ã ɒ̃ ɓ̃ ɖ̃ ɗ̃ ə̃ ə̃ ɛ̃ ɜ̃ ɘ̃ ɟ̃ ɡ̃ ɡ̃ ɢ̃ ɣ̃ ɤ̃ ɥ̃ ɦ̃ ɨ̃ ɨ̃ ɩ̃ ɩ̃ ɪ̃ ɪ̃ ɭ̃ ɲ̃ ɴ̃ ɵ̃ œ̃ ɾ̃ ɾ̃ ʂ̃ ʃ̃ ʊ̃ ʌ̃ ʏ̃ ʏ̃ ʐ̃ ʑ̃ ʒ̃ ʝ̃ ɮ̃
ā ƀ ɔ̄ đ̄ ē f̄ ḡ ī j̄ k̄ l̄ m̄ n̄ ō p̄ q̄ r̄ s̄ t̄ ū v̄ w̄ x̄ ȳ z̄ ǣ œ̄ ŋ̄ ɐ̄ ā ɒ̄ ɓ̄ ɖ̄ ɗ̄ ə̄ ə̄ ɛ̄ ɜ̄ ɘ̄ ɟ̄ ɡ̄ ɡ̄ ɢ̄ ɣ̄ ɤ̄ ɥ̄ ɦ̄ ɨ̄ ɨ̄ ɩ̄ ɩ̄ ɪ̄ ɪ̄ ɭ̄ ɲ̄ ɴ̄ ɵ̄ œ̄ ɾ̄ ɾ̄ ʂ̄ ʃ̄ ʊ̄ ʌ̄ ʏ̄ ʏ̄ ʐ̄ ʑ̄ ʒ̄ ʝ̄ ɮ̄
ȧ ƀ ɔ̇ đ̇ ė ḟ ġ ṁ ṅ ȯ ṗ q̇ ṙ ṡ ṫ u̇ v̇ ẇ ẋ ẏ ż ǽ œ̇ ŋ̇ ɐ̇ ȧ ɒ̇ ɓ̇ ɖ̇ ɗ̇ ə̇ ə̇ ɛ̇ ɜ̇ ɘ̇ ɟ̇ ɡ̇ ɡ̇ ɢ̇ ɣ̇ ɤ̇ ɥ̇ ɦ̇ ɨ̇ ɨ̇ ɩ̇ ɩ̇ ɪ̇ ɪ̇ ɭ̇ ɲ̇ ɴ̇ ɵ̇ œ̇ ɾ̇ ɾ̇ ʂ̇ ʃ̇ ʊ̇ ʌ̇ ʏ̇ ʏ̇ ʐ̇ ʑ̇ ʒ̇ ʝ̇ ɮ̇
ḁ ḃ ḉ ḋ ḗ ḟ ḡ ḭ ḱ ḷ ṁ ṇ ṓ ṗ q̇ ṛ ṡ ṭ ṵ v̇ ẇ ẋ ẏ ẓ ǽ œ̇ ŋ̇ ɐ̇ ȧ ɒ̇ ɓ̇ ɖ̇ ɗ̇ ə̇ ə̇ ɛ̇ ɜ̇ ɘ̇ ɟ̇ ɡ̇ ɡ̇ ɢ̇ ɣ̇ ɤ̇ ɥ̇ ɦ̇ ɨ̇ ɨ̇ ɩ̇ ɩ̇ ɪ̇ ɪ̇ ɭ̇ ɲ̇ ɴ̇ ɵ̇ œ̇ ɾ̇ ɾ̇ ʂ̇ ʃ̇ ʊ̇ ʌ̇ ʏ̇ ʏ̇ ʐ̇ ʑ̇ ʒ̇ ʝ̇ ɮ̇
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
ạ ḃ ọ ḍ ẹ ḟ ḡ ị ḳ ḷ ṃ ṇ ọ ṗ q̣ ṛ ṣ ṭ ụ ṿ ẉ ẋ ỵ ẓ ǽ œ̣ ŋ̣ ɐ̣ ạ ɒ̣ ɓ̣ ɖ̣ ɗ̣ ə̣ ə̣ ɛ̣ ɜ̣ ɘ̣ ɟ̣ ɡ̣ ɡ̣ ɢ̣ ɣ̣ ɤ̣ ɥ̣ ɦ̣ ɨ̣ ɨ̣ ɩ̣ ɩ̣ ɪ̣ ɪ̣ ɭ̣ ɲ̣ ɴ̣ ɵ̣ œ̣ ɾ̣ ɾ̣ ʂ̣ ʃ̣ ʊ̣ ʌ̣ ʏ̣ ʏ̣ ʐ̣ ʑ̣ ʒ̣ ʝ̣ ɮ̣
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ a ɒ ɓ ɖ ɗ ə ə ɛ ɜ ɘ ɟ ɡ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɨ ɩ ɩ ɪ ɪ ɭ ɲ ɴ ɵ œ ɾ ɾ ʂ ʃ ʊ ʌ ʏ ʏ ʐ ʑ ʒ ʝ ɮ
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ a ɒ ɓ ɖ ɗ ə ə ɛ ɜ ɘ ɟ ɡ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɨ ɩ ɩ ɪ ɪ ɭ ɲ ɴ ɵ œ ɾ ɾ ʂ ʃ ʊ ʌ ʏ ʏ ʐ ʑ ʒ ʝ ɮ
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ŋ ɐ a ɒ ɓ ɖ ɗ ə ə ɛ ɜ ɘ ɟ ɡ ɡ ɢ ɣ ɤ ɥ ɦ ɨ ɨ ɩ ɩ ɪ ɪ ɭ ɲ ɴ ɵ œ ɾ ɾ ʂ ʃ ʊ ʌ ʏ ʏ ʐ ʑ ʒ ʝ ɮ

Regular, Bases requiring anchors for rare marks, Rare combining marks

Regular patched, Bases requiring anchors for rare marks, Rare combining marks

Italic, Bases requiring anchors for rare marks, Rare combining marks

Italic patched, Bases requiring anchors for rare marks, Rare combining marks

Semibold, Bases requiring anchors for rare marks, Rare combining marks

Semibold patched, Bases requiring anchors for rare marks, Rare combining marks

Semibold italic, Bases requiring anchors for rare marks, Rare combining marks

Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks

Patch superscript consonant anchors

Libertinus serif regular

h ɦ j ɾ ɹ ɻ ʁ w y ɣ l s x ʕ b k m p t ɯ v ɐ β ɣ ð

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵊ ᵝ ᵞ ᵟ

ʰ ʱ ʲ ʳ ʴ ʵ ʷ ʸ ˡ ˢ ˣ ˤ ᵇ ᵏ ᵐ ᵖ ᵗ ᵚ ᵛ ᵊ ᵝ ᵞ ᵟ

h ɦ j ɾ ɹ ɻ ʁ w y ɣ l s x ʕ b k m p t ɯ v ɐ β ɣ ð

ɦ ɦ j ɾ ɹ ɻ ʁ ẁ ỳ ɣ̀ l̀ s̀ x̀ ʕ̀ b̀ k̀ m̀ p̀ t̀ ɯ̀ v̀ ɐ̀ β̀ ɣ̀ ð̀

ɦ ɦ j ɾ ɹ ɻ ʁ ẃ ý ɣ́ ĺ ś x́ ʕ́ b́ ḱ ḿ ṕ t́ ɯ́ v́ ɐ́ β́ ɣ́ ð́

ɦ ɦ j ɾ ɹ ɻ ʁ ŵ ŷ ɣ̂ l̂ ŝ x̂ ʕ̂ b̂ k̂ m̂ p̂ t̂ ɯ̂ v̂ ɐ̂ β̂ ɣ̂ ð̂

Libertinus serif regular, patched

h'ɦ'j'ɾ'ɹ'ɻ'ʁ'w'y'ɣ'l'ˢ'ˣ'ˤ'ᵇ'ᵏ'ᵐ'ᵖ'ᵗ'ᵚ'ᵛ'ᵊ'ᵝ'ᵞ'ᵟ'

h'ɦ'j'ɾ'ɹ'ɻ'ʁ'w'y'ɣ'l'ˢ'ˣ'ˤ'ᵇ'ᵏ'ᵐ'ᵖ'ᵗ'ᵚ'ᵛ'ᵊ'ᵝ'ᵞ'ᵟ'

h'ɦ'j'ɾ'ɹ'ɻ'ʁ'w'y'ɣ'l'ˢ'ˣ'ˤ'ᵇ'ᵏ'ᵐ'ᵖ'ᵗ'ᵚ'ᵛ'ᵊ'ᵝ'ᵞ'ᵟ'

h̀ ɦ̀ j̀ ɾ̀ ɹ̀ ɻ̀ ʁ̀ ẁ ỳ ɣ̀ l̀ s̀ x̀ ʕ̀ b̀ k̀ m̀ p̀ t̀ ɯ̀ v̀ ɐ̀ β̀ ɣ̀ ð̀

ɦ́ ɦ́ j́ ɾ́ ɹ́ ɻ́ ʁ́ ẃ ý ɣ́ ĺ ś x́ ʕ́ b́ ḱ ḿ ṕ t́ ɯ́ v́ ɐ́ β́ ɣ́ ð́

ɦ̂ ɦ̂ ĵ ɾ̂ ɹ̂ ɻ̂ ʁ̂ ŵ ŷ ɣ̂ l̂ ŝ x̂ ʕ̂ b̂ k̂ m̂ p̂ t̂ ɯ̂ v̂ ɐ̂ β̂ ɣ̂ ð̂

Gentium Plus

hʰ ɦʱ jʲ rʳ ɹʴ ɻʵ ʁʷ wʷ yʸ ɣʸ lˡ sˢ xˣ ʕˤ bᵇ kᵏ mᵐ pᵖ tᵗ ɯᵚ vᵛ ɐᵊ βᵝ ɣᵞ ðᵟ

hʰ ɦʱ jʲ rʳ ɹʴ ɻʵ ʁʷ wʷ yʸ ɣʸ lˡ sˢ xˣ ʕˤ bᵇ kᵏ mᵐ pᵖ tᵗ ɯᵚ vᵛ ɐᵊ βᵝ ɣᵞ ðᵟ

hᵐ ɦᵐ jᵐ rᵐ ɹᵐ ɻᵐ ʁᵐ wᵐ yᵐ ɣᵐ lᵐ sᵐ xᵐ ʕᵐ bᵐ kᵐ mᵐ pᵐ tᵐ ɯᵐ vᵐ ɐᵐ βᵐ ɣᵐ ðᵐ

h̀ ɦ̀ j̀ ɾ̀ ɹ̀ ɻ̀ ʁ̀ ẁ ỳ ɣ̀ l̀ s̀ x̀ ʕ̀ b̀ k̀ m̀ p̀ t̀ ɯ̀ v̀ ɐ̀ β̀ ɣ̀ ð̀

ɦ́ ɦ́ j́ ɾ́ ɹ́ ɻ́ ʁ́ ẃ ý ɣ́ ĺ ś x́ ʕ́ b́ ḱ ḿ ṕ t́ ɯ́ v́ ɐ́ β́ ɣ́ ð́

ɦ̂ ɦ̂ ĵ ɾ̂ ɹ̂ ɻ̂ ʁ̂ ŵ ŷ ɣ̂ l̂ ŝ x̂ ʕ̂ b̂ k̂ m̂ p̂ t̂ ɯ̂ v̂ ɐ̂ β̂ ɣ̂ ð̂

Aᵐ Bᵐ Cᵐ Dᵐ Eᵐ Fᵐ Gᵐ Hᵐ Iᵐ Jᵐ Kᵐ Lᵐ Mᵐ Nᵐ Oᵐ Pᵐ Qᵐ Rᵐ Sᵐ Tᵐ Uᵐ Vᵐ Wᵐ Xᵐ Yᵐ Zᵐ aᵐ bᵐ cᵐ dᵐ eᵐ fᵐ gᵐ hᵐ iᵐ jᵐ kᵐ lᵐ mᵐ nᵐ oᵐ pᵐ qᵐ rᵐ sᵐ tᵐ uᵐ vᵐ wᵐ xᵐ yᵐ zᵐ Æᵐ æᵐ Œᵐ œᵐ Øᵐ øᵐ Ðᵐ ðᵐ Þᵐ þᵐ ßᵐ ʃᵐ Đᵐ đᵐ Ħᵐ ħᵐ ɪᵐ jᵐ Łᵐ łᵐ Ƀᵐ Бᵐ

À B̀ Ć D̀ È F̀ Ǵ H̀ Ì J̀ Ḱ L̀ M̀ Ń Ò P̀ Q́ R̀ S̀ T̀ Ù V̀ Ẁ X̀ Ỳ Ż à b̀ ć d̀ è f̀ ǵ ḥ ì j̀ ḱ l̀ ḿ ń ò p̀ q́ ŕ s̀ t̀ ù v̀ ẁ x̀ ỳ ż Æ̀ æ̀ Œ̀ œ̀ Ø̀ ø̀ Ð̀ ð̀ Þ̀ þ̀ ß̀ ſ́ Đ̀ đ̀ Ħ̀ ħ̀ ì j̀ Ł́ ł́ Ƀ̀ Б̀